Question 1:
The marks obtained by a student in 5 different subjects are input by the user. The student gets a division as per the following rules:
Percentage above or equal to 60 - First division
Percentage between 50 and 59 - Second division
Percentage between 40 and 49 - Third division
Percentage less than 40 - Fail
Write a program to calculate the division obtained by the student.

Question 2:
Write a program to compute sin x for given x. The user should supply x and a positive integer n. We compute the sine of x using the series and the computation should use all terms in the series up through the term involving xn
sin x = x - x3/3! + x5/5! - x7/7! + x9/9! ....

Question 3:
Write a recursive function to print all permutations of a given string.

Question 4:
Write a recursive function to generate all possible subsets of a set.

Question 5:
Write a menu-driven program that performs arithmetic operations (addition, subtraction, multiplication, division) based on user input.

Question 6:
In a company an employee is paid as under:
If his basic salary is less than Rs. 1500, then HRA = 10% of basic salary and DA = 90% of basic salary.
If his salary is either equal to or above Rs. 1500, then HRA = Rs. 500 and DA = 98% of basic salary.
If the employee's salary is input by the user, write a program to find his gross salary.

Question 7:
Suppose A, B, C are arrays of integers of size M, N, and M + N respectively. The numbers in array A appear in ascending order while the numbers in array B appear in

descending order. Write a user defined function in C++ to produce third array C by merging arrays A and B in ascending order. Use A, B and C as arguments in the function.

Question 8:

Write a program to initialize 3 different strings and perform the following functions:

1. Copy STRING1 into STRING3
2. Concatenate STRING1 and STRING2
3. Calculate and show the total length of STRING1 after concatenation

Question 9:

Write a program that takes inputs from the user within a described range, and then adds only the odd numbers within the array.

Question 10:

Write a program to rotate an array by a given number of positions.

Question 11:

Write a function that takes an array of numbers and returns an array with two elements:

1. The first element should be the sum of all even numbers in the array.
2. The second element should be the sum of all odd numbers in the array.

Question 12:

Write a program to find the union and intersection of two arrays.

Question 13:

Write a program that transposes a 2D matrix.

Question 14:

Write a menu-driven program to perform various operations on an array, such as insert, delete, search, and display, using loops and control flow.

Question 15:

Write a function named "digit_name" that takes an integer argument in the range from 1 to 9 , inclusive, and prints the English name for that integer on the computer screen. No newline character should be sent to the screen following the digit name. The function should not return a value.

Question 16:

Write a function named "reduce" that takes two positive integer arguments, call them "num" and "denom", treats them as the numerator and denominator of a fraction, and reduces the fraction. That is to say, each of the two arguments will be modified by dividing it by the greatest common divisor of the two integers. The function should return the value 0 (to indicate failure to reduce) if either of the two arguments is zero or negative, and should return the value 1 otherwise.